The Ghost: an ephemeral inner-reasoning layer

Concept note and working prototype — v0.01

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Local draft, unpublished — v0.01

Abstract

The Ghost is an inner-reasoning layer that activates before output generation, produces a thought in a private numeric code, uses it to reformulate the request, and then erases it. The user only sees the final output. This note collects the original intuition (Catania, October 7 2026), three design decisions that make it implementable, the `src/fantasma` prototype, and the first live experiments with DeepSeek-V4.1-Flash (API id `deepseek-flash`). Four real-world cases and a code test show the Ghost excels at framing and reformulating, while execution remains a second step. The trade-off between cognitive privacy and verifiability is structural and is made explicit, not hidden.

1 Idea

The Ghost emerges, acts, disappears. It leaves a trace only for whoever evoked it. Unlike visible Chain-of-Thought, hidden scratchpads and latent reasoning, erasure here is active and the code is private: a numeric sequence (e.g. `0984567345679`) uninterpretable from the outside.

2 Mechanism

1. Input reaches the model.
2. An internal gate decides whether to activate the Ghost.
3. The Ghost generates a private numeric thought.
4. The same model re-reads its own code (decision 5.1).
5. The reformulated input produces the final output.
6. The thought is erased; only a cryptographic commitment (hash) remains.

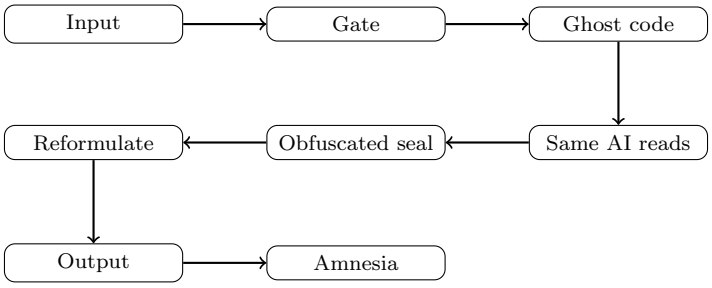


Figure 1: The Ghost cycle: emerge, act, disappear. Only the hash audit remains.

3 Three design decisions

5.1 — The same AI interprets. No external decoder: the code is handed back to the same model via a `FANTASMA-INTERPRETA` prompt. Only that state, in that turn, can read it. Without salt and nonce it cannot be reconstructed.

5.2 — Obfuscated cache. The raw code lives in ephemeral memory only for the interpretation turn. It is then sealed into cache in obfuscated form (digit shift derived from the salt, i.e. the AI's inner state). The cache never holds the raw; with the wrong salt one gets noise.

5.3 — Amnesia after output. Once the output is out, there is nothing to remember. Ephemeral memory and cache are wiped. No persistent vector store. Only the hash-chained audit remains, recording commitments, never contents.

4 Prototype

`src/fantasma`: `gate.py` (inspectable heuristic gate), `codec.py` (HMAC + nonce, every Ghost unique), `memory.py` (`EphemeralMemory` + `ObfuscatedCache`), `audit.py` (content-free hash chain), `pipeline.py` (1–8 flow), `llm_deepseek.py` (DeepSeek-V4.1-Flash adapter, key from env only). Suite: 12 tests green locally and on Docker (`docker run --rm fantasma:0.1`).

5 Live experiments (09/10/2026)

Model: DeepSeek-V4.1-Flash (`deepseek-flash`), `reasoning_effort=high`, `thinking=enabled`. All cases: ghost on, same-AI on, memory and cache empty afterwards, audit verified.

Case	Reformulation	Output
Milan/Catania decision	values and constraints made explicit	clarified question
Message to collaborator	firm-professional tone	ready-to-send draft
Web-app brief	phases, alternatives, risks	operational question
Ethical paradox	procedural conditions	thesis: consent and contestability
Code test	4 bugs identified	analysis + fix in 2 steps

On code the Ghost found every hidden flaw (mutable default, `range(len+1)`, division by zero, misleading names), produced an executable specification and, at the second step, 391 lines of implementation with tests. First run red on one over-strict assert (extra keys), then green after a one-line fix: `Tutti i test sono passati.`

6 Open problems and the ethical paradox

What remains: grounding a private language (Wittgenstein), causal erasure being impossible in a transformer, learning without persistent memory, auditing without revealing. The principle "no one shall decide how someone else reasons" stays in tension with gate, code and erasure being designed: the gate is therefore explicit and configurable, never hidden. The privacy–verifiability trade-off is structural.

7 Vision

The Ghost does not aim at a more opaque model, but at a legitimate, procedural opacity: consent, contestability, time bounds, independent gate oversight.

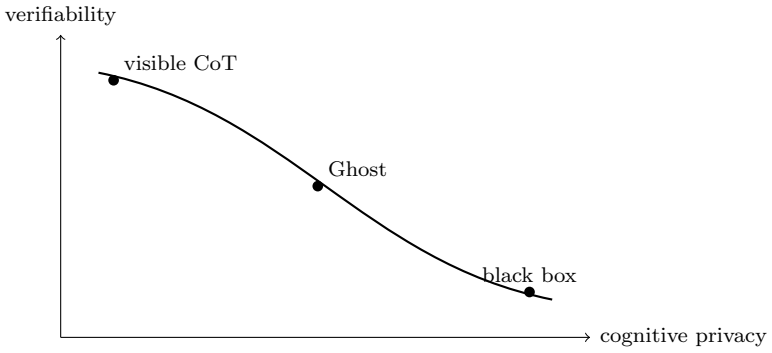


Figure 2: The more private the thought, the less verifiable. The Ghost sits in the middle, declared.

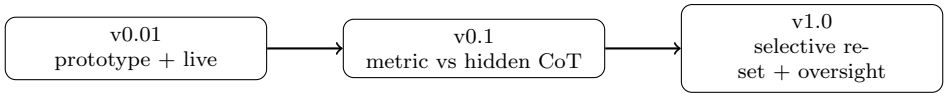


Figure 3: Roadmap: from a working prototype to measurement, then to guarantees.

8 Status and next steps

Status: concept + working v0.01 prototype, 12 green tests, recorded live experiments. Next: measurable improvement metric vs. hidden CoT, true selective reset, independent gate oversight. Otherwise it stays in the notebook: not every idea must become a paper; some are just beautiful. This one, for now, works.